Open problems:
Physical vector spaces for quantum mechanics

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

From the brief

1. **Completeness of the Lie algebra.** Definition 1 admits pathological cases: a Hilbert space with the defining observables of a qubit ("too many vectors") and a Schwartz space with position as the only defining observable ("too few observables"). "Addressing this gap is an open problem"; the brief expects a "final definition" once it is solved. (Brief, Sec. 5, "Completeness of the Lie algebra"; Sec. 6.)

2. **Isomorphisms of quantum vector spaces.** The brief speaks of isomorphisms "in the category of quantum vector spaces" that "must preserve the Lie algebraic structure", but does not define the morphisms. (Brief, Example 3; Theorem 5, proof; Log, "Differences", last item.)

3. **$\mathcal{S}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^m)$ iff $n = m$.** Stated in Example 3 and argued only by "no linear transformation can change the number of defining observables", although the number of defining observables of a presentation is not shown to be an invariant. Not formalized. (Brief, Example 3 and Remark; Log, "Differences", last item.)

4. **Schwartz spaces as topological vector spaces.** "Schwartz spaces of different dimensionality are isomorphic as topological vector spaces" is stated without proof or reference. (Brief, Remark after Example 3.)

5. **Lie algebras determine physical systems.** "Two systems are physically equivalent if and only if their Lie algebras of observables are equivalent" is asserted, and is what makes QSTO:V consistent with [B]STDIST. (Brief, Theorem 5, proof; Log, "Differences", last item.)

6. **Frame changes.** The [B]WDQ argument assumes that the defining observables X_i^A of one frame are continuous everywhere-defined observables in the other frame, that is, that a change of frame is an isomorphism of quantum vector spaces. This is not shown. (Brief, Theorem 5, proof; Log, "Differences", last item.)

7. **Ensembles on a quantum vector space.** QSTO:V uses density operators ρ and $S(\rho) = -\operatorname{tr}(\rho \log \rho)$, and Sec. 5 calls the set of ensembles "the most natural space to consider", but no ensemble space over V is defined. (Brief, Sec. 3, QSTO:V; Sec. 5.)

8. **Second countability from the pure states.** "If the pure states are second countable, we can always extend the topology to the vector space to preserve second countability" is stated without proof. (Brief, Sec. 5, "Second countability vs separability".)

9. **A countable generating set of observables.** "Separability/second countability would allow us to find a countable subset of observables from a bigger set that generates the same topology" is stated without proof. (Brief, Sec. 5, "Number of defining observables".)

10. **Field theories.** "For a field theory, we would expect the defining observables to be at least countable"; no field-theory example of a quantum vector space is given. (Brief, Sec. 5, "Number of defining observables".)

11. **Which observables generate unitary groups.** Since symmetric and self-adjoint coincide on V, "not all self-adjoint operators generate a unitary transformation", and "to understand which ones the integrability of the system must be checked". A characterization, analogous to complete vector fields in Hamiltonian mechanics, is not given. (Brief, Sec. 5, "Symmetric vs self-adjoint".)

From the formalization

1. [BLOCKS VERIFICATION: FORMALIZATION GAP] **Completeness and second countability.** The conditions "complete" and "second-countable" of Definition 1 are not formalized, so Example 3 is not checked to be a quantum vector space; Mathlib has no completeness of $\mathcal{S}$ at this pin. (Log, "Differences", last item.)

2. [BLOCKS VERIFICATION: FORMALIZATION GAP] **Example 3 in n dimensions.** Example 3 and the Remark after it (the topology is that of $\mathcal{S}$) are formalized for $\mathcal{S}(\mathbb{R})$ only, not for $\mathcal{S}(\mathbb{R}^n)$ with observables $\{X^i, P_i, I\}$. (Log, "Differences".)